ACADEMIC REGULATIONS
AND
COURSE STRUCTURE

CHOICE BASED CREDIT SYSTEM

R22

Computer Science & Engineering

(DATA SCIENCE)

Bachelor of Technology (B.Tech)

B. Tech. - Regular Four-Year Degree Programme

(For batches admitted from the academic year 2022-2023)

&

(For batches admitted Lateral Entry Scheme from the academic year 2023-2024)

MLR Institute of Technology

(Autonomous)

Laxman Reddy Avenue, Dundigal (V), Quthbullapur (M),
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COURSE STRUCTURE

Course Structure
B. TECH – Computer Science and Engineering (Data Science)
Regulations: R22

I B.Tech.- I Semester

Induction program for one weeks

Code	Course	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A6BS01	Linear Algebra and Calculus	BSC	3	1	0	4	40	60	100
A6CS02	Programming for Problem Solving	ESC	3	0	0	3	40	60	100
A6HS01	English for Skill Enhancement	HSMC	3	0	0	3	40	60	100
A6BS09	Engineering Chemistry	BSC	3	1	0	4	40	60	100
A6EC03	Electronic Devices and Applications	ESC	2	0	0	2	40	60	100
A6CS03	Programming for Problem Solving Lab	ESC	0	0	3	1.5	40	60	100
A6HS02	English Language and Communication Skills Lab	HSMC	0	0	3	1.5	40	60	100
A6HS04	Seminar	HSMC	0	0	2	1	50	-	50
A6BS11	Environmental Science	MC	3	0	0	-	50	-	50
		TOTAL	**17**	**2**	**8**	**20**	**380**	**420**	**800**

I B.Tech.- II Semester

Code	Course	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A6BS02	Numerical Methods and Integral Transforms	BSC	3	1	0	4	40	60	100
A6BS07	Applied Physics	BSC	3	1	0	4	40	60	100
A6EE60	Basic Electrical and Electronics Engineering	BSC	3	1	0	4	40	60	100
A6ME02	Engineering Drawing	ESC	1	0	3	2.5	40	60	100
A6BS08	Applied Physics Lab	BSC	0	0	3	1.5	40	60	100
A6CS04	Python Programming Lab	ESC	0	0	3	1.5	40	60	100
A6ME04	Engineering Work Shop	ESC	0	0	3	1.5	40	60	100
A6DS01	Data Handling and Interpretation	ESC	0	1	2	1	50	-	50
		TOTAL	**10**	**4**	**14**	**20**	**330**	**420**	**750**

II B.Tech.- I Semester										
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A6CS08	Discrete Mathematics	BSC	3	0	0	3	40	60	100
A6CS11	Operating Systems	PCC	3	0	0	3	40	60	100
A6CS28	Digital Electronics and Computer Organization	ESC	3	0	0	3	40	60	100
A6CS09	Database Management Systems	PCC	3	0	0	3	40	60	100
A6CS05	Data structures	PCC	3	0	0	3	40	60	100
A6CS10	Database Management Systems Lab	PCC	0	0	3	1.5	40	60	100
A6CS06	Data Structures Lab	PCC	0	0	3	1.5	40	60	100
A6DS04	Skill Development Course (Data Wrangling)	PCC	0	0	4	2	100	-	100
A6HS06	Constitution of India	MC	2	-	-	0	50	-	50
Total			15	00	12	20	420	480	900

II B.Tech.- II Semester									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A6BS03	Computer Oriented Statistical Methods	BSC	3	1	0	4	40	60	100
A6HS08	Business Economics and Financial Analysis	HSC	3	0	0	3	40	60	100
A6DS02	Data Analytics using R	PCC	2	0	0	2	40	60	100
A6IT02	Object Oriented Programming through Java	ESC	3	0	0	3	40	60	100
A6CS15	Design and Analysis of Algorithms	PCC	3	0	0	3	40	60	100
A6DS03	Data Analytics using R Lab	PCC	0	0	3	1.5	40	60	100
A6IT03	Object Oriented Programming through Java Lab	ESC	0	0	3	1.5	40	60	100
A6DS05	Real-time research project / Societal Related Project	PWC	0	0	4	2	50	-	50
A6HS05	Gender Sensitization	MC	2	-	-	0	50	-	50
Total			14	01	10	20	360	540	900

III B.Tech.- I Semester										
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A6DS06	Big Data Technologies	PCC	3	0	0	3	40	60	100
A6IT11	Automata and Compiler Design	PCC	3	0	0	3	40	60	100
A6AI07	Web Programming	PCC	3	0	0	3	40	60	100
	Professional Elective-I	PEC	3	0	0	3	40	60	100
A6CS07	Software Engineering	PCC	3	0	0	3	40	60	100
A6AI08	Web Programming Lab	PCC	0	0	3	1.5	40	60	100
A6DS07	Big Data Technologies Lab	PCC	0	0	3	1.5	40	60	100
A6DS12	MOOCS/Independent Study	PCC	0	0	0	2	40	60	100
A6HS10	Human Values and Professional Ethics	MC	2	-	-	-	50	-	50
Total			15	00	08	20	360	540	900

III B.Tech.- II Semester									
Course Code	Course Title	Cours e Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A6DS31	Stream Processing and Analytics	PCC	3	1	0	4	40	60	100
A6CS07	Computer Networks	PCC	3	0	0	3	40	60	100
	Professional Elective-II	PEC	3	0	0	3	40	60	100
	Professional Elective-III	PEC	3	0	0	3	40	60	100
	Open Elective-I	OEC	3	0	0	3	40	60	100
A6DS32	Stream Processing and Analytics Lab	PCC	0	0	3	1.5	40	60	100
	Professional Elective-II Lab	PEC	0	0	3	1.5	40	60	100
A6DS35	Industrial Oriented Mini Project	PWC	0	0	3	1		100	100
A6BS11	Environmental Science	MC	2	-	-	0	50	-	-
Total			15	01	08	20	320	480	800

IV B.Tech.- I Semester									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A6A117	Deep Learning	PCC	3	0	0	3	40	60	100
A6DS15	Web and Social Media Analytics	PCC	3	0	0	3	40	60	100
	Professional Elective-IV	PEC	3	0	0	3	40	60	100
	Professional Elective-V	PEC	3	0	0	3	40	60	100
	Open Elective-II	OEC	3	0	0	3	40	60	100
A6A119	Deep Learning Lab	PCC	0	0	2	1	40	60	100
A6DS29	Research Project Phase - 1	PCC	0	0	8	4	100	-	100
Total			15	00	10	20	280	420	700

IV B.Tech. - II Semester									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A6HS15	Organizational Behavior	HSMC	3	0	0	3	40	60	100
	Professional Elective-VI	PEC	3	0	0	3	40	60	100
	Open Elective - III	OEC	3	0	0	3	40	60	100
A6DS30	Research Project Phase – 2	PCC	0	0	16	9 + 2	40	60	100
Total			09	00	16	20	160	240	400

SEM	I	II	III	IV	V	VI	VII	VIII	**Total**
Credits	20	20	20	20	20	20	20	20	**160**

PROFESSIONAL ELECTIVES			
PE - I		**PE – II**	
A6AI11	Data Mining Techniques	**A6DS32**	**High Dimensional Data Analysis**
A6DS09	Exploratory Data Analytics	**A6DS23**	Internet Of Things (IOT)
A6DS10	Data Integration	**A6DS13**	Data Wrangling
A6DS11	Python for Data Science	**A6DS25**	Object Oriented Analysis and Design (OOAD)
PE - III		**PE - IV**	
A6DS29	Cloud Computing	**A6AI14**	Natural Language Processing
A6DS28	Predictive Analytics	**A6DS16**	Data Base Security
A6IT13	Cloud & DevOps	**A6DS17**	Time Series Analysis and Forecasting
A6IT40	Introduction to Artificial Intelligence	**A6DS18**	Edge Analytics
PE - V		**PE - VI**	
A61T26	Information Retrieval Systems	**A6DS21**	Data Visualization using Tableau
A6DS19	Video Analytics	**A6DS22**	Business Intelligence and Analytics
A6CY25	Block Chain Technology	**A6IT10**	Full Stack Development
A6DS20	Data Collection and Analysis with IOT	**A6DS27**	Big Query Systems
PE – II LAB			
A6DS33	**High Dimensional Data Analysis Lab**		
A6DS24	Internet Of Things (IOT) Lab		
A6DS14	Data Wrangling Lab		
A6DS26	Object Oriented Analysis and Design (OOAD) Lab		